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(54) **ELECTRIC POWER STEERING DEVICE**

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(71) Applicant: **Mitsubishi Electric Corporation**,
Tokyo (JP)

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(72) Inventors: **Akio Numakura**, Tokyo (JP); **Hideki Doi**, Tokyo (JP)

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(73) Assignee: **Mitsubishi Electric Corporation**,
Tokyo (JP)

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Primary Examiner — Joan T Goodbody

Assistant Examiner — Godfrey Aleksander Maciorowski

(74) Attorney, Agent, or Firm — Sughrue Mion, PLLC

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(57) **ABSTRACT**

This electric power steering device is provided with a function for: notifying a parking assistance device of a statically-steerable upper limit number obtained from an estimated motor current when static steering is performed in turnabout and a current limit value for the purpose of overheat protection of an electric power steering; and notifying the parking assistance device of whether the electric power steering can perform steering according to a target steering angle command, from a target steering angle and a target vehicle speed from the start of parking to the completion thereof, thus preventing interruption due to overheat of the electric power steering during parking assistance.

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B62D 15/02 (2006.01)

(52) **U.S. Cl.**

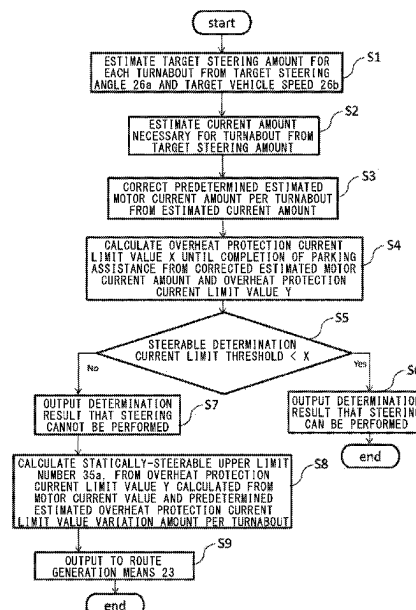
CPC **B62D 5/046** (2013.01); **B62D 5/0496** (2013.01); **B62D 15/0285** (2013.01)

(58) **Field of Classification Search**

CPC ... B62D 5/046; B62D 5/0496; B62D 15/0285

See application file for complete search history.

7 Claims, 7 Drawing Sheets



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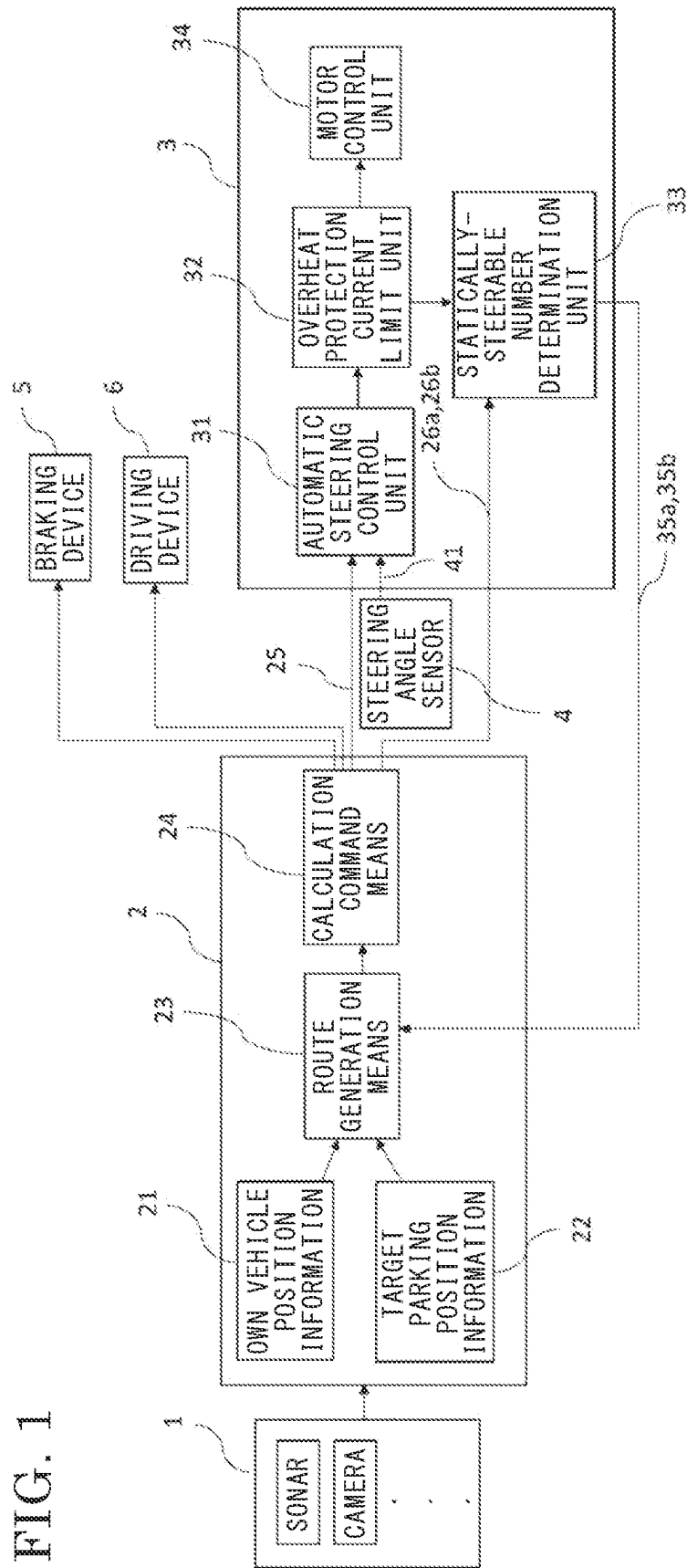
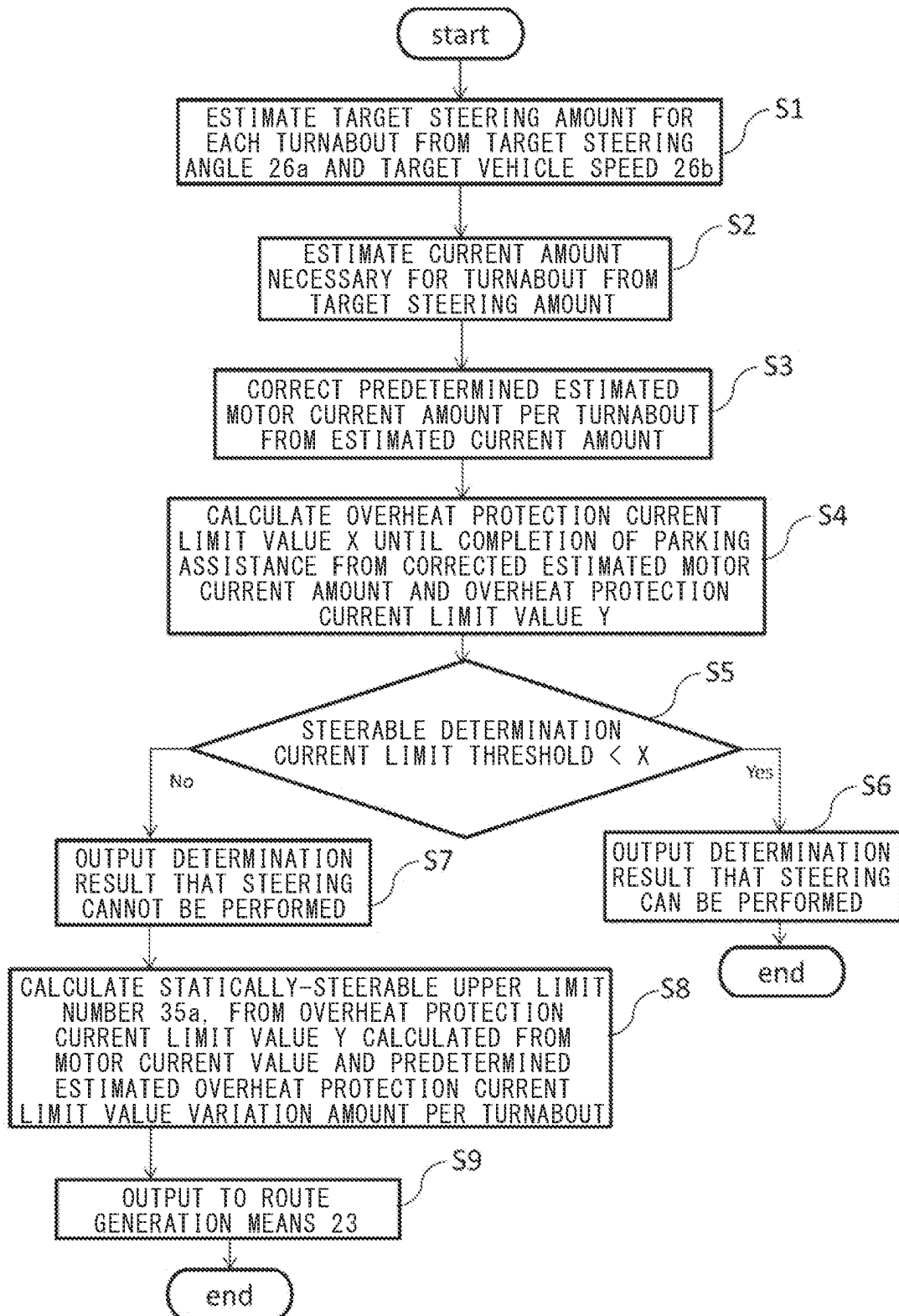


FIG. 2



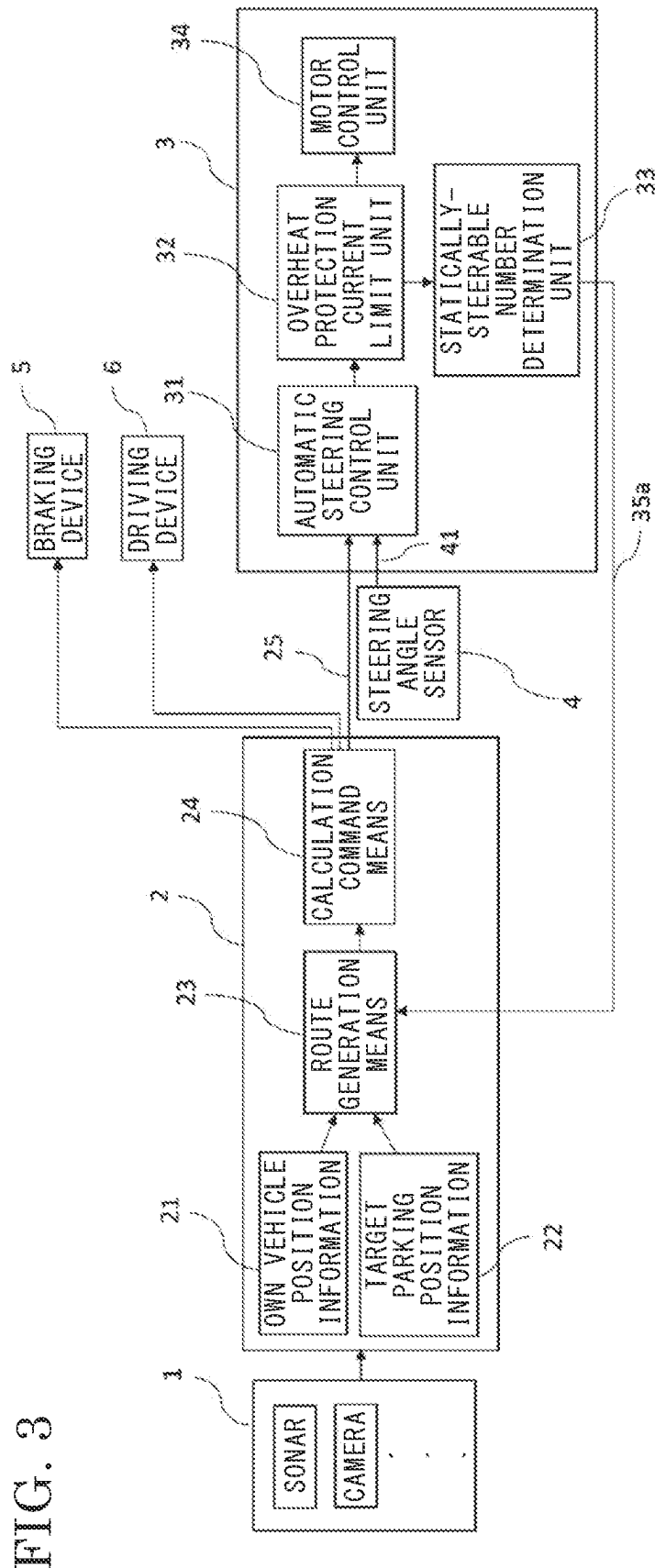


FIG. 4

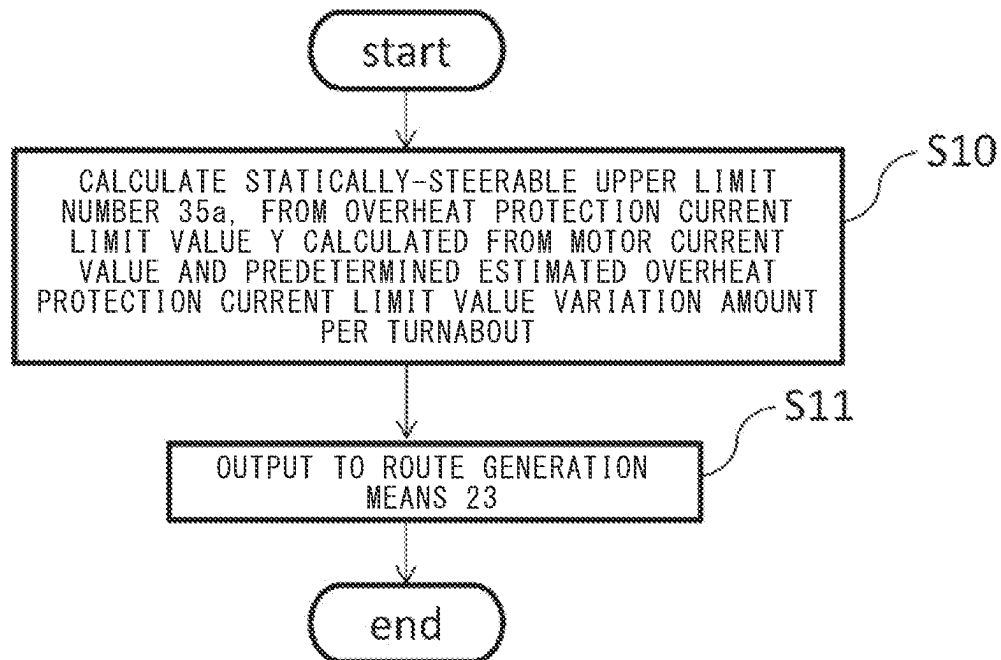


FIG. 5

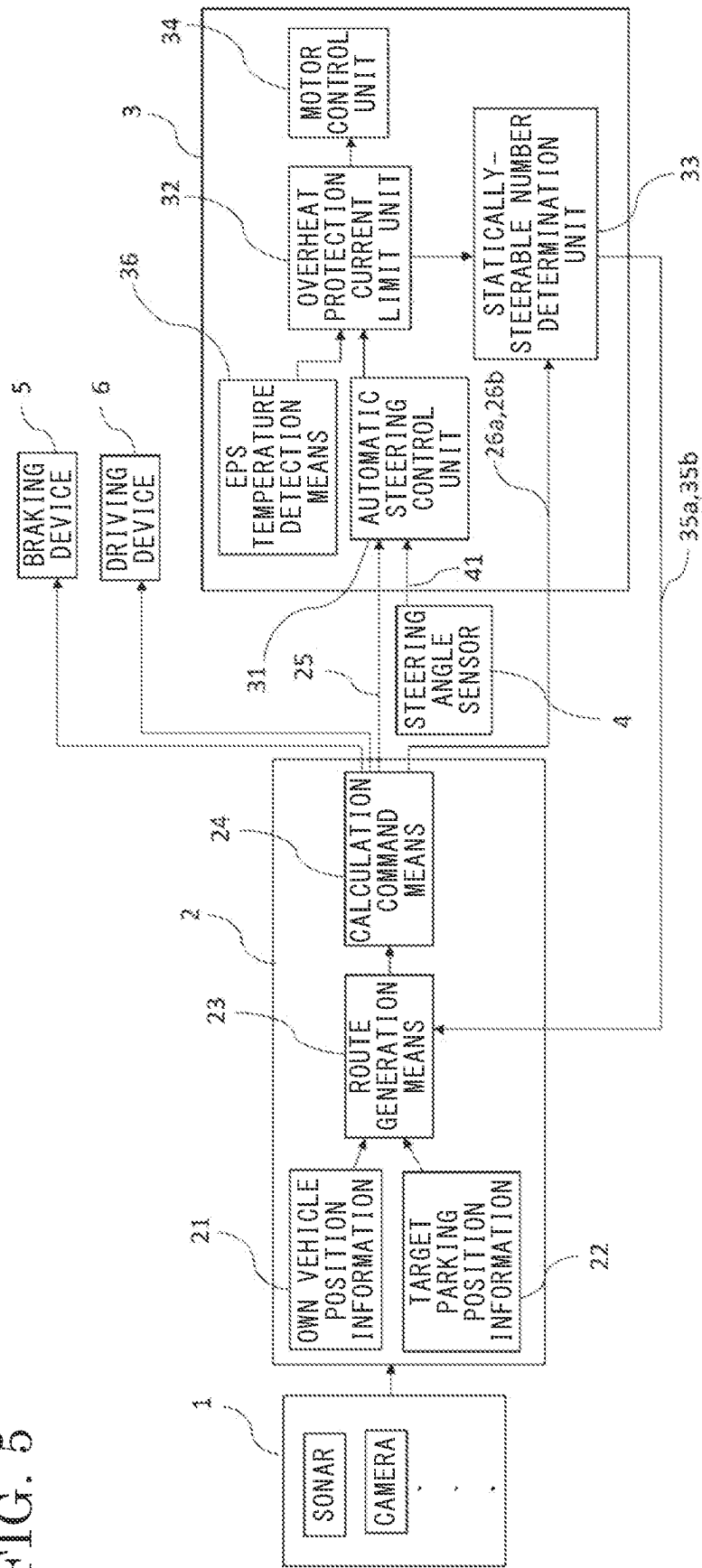


FIG. 6

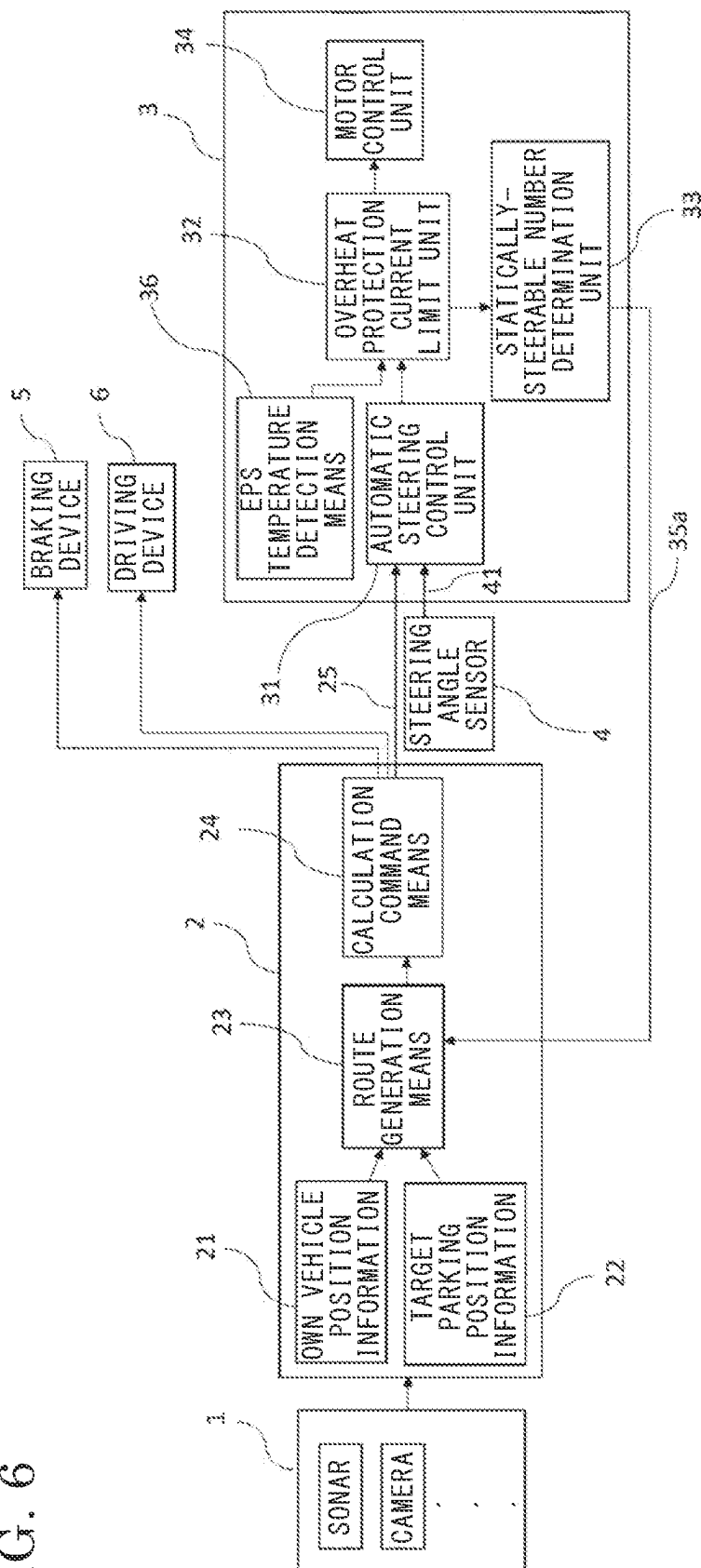
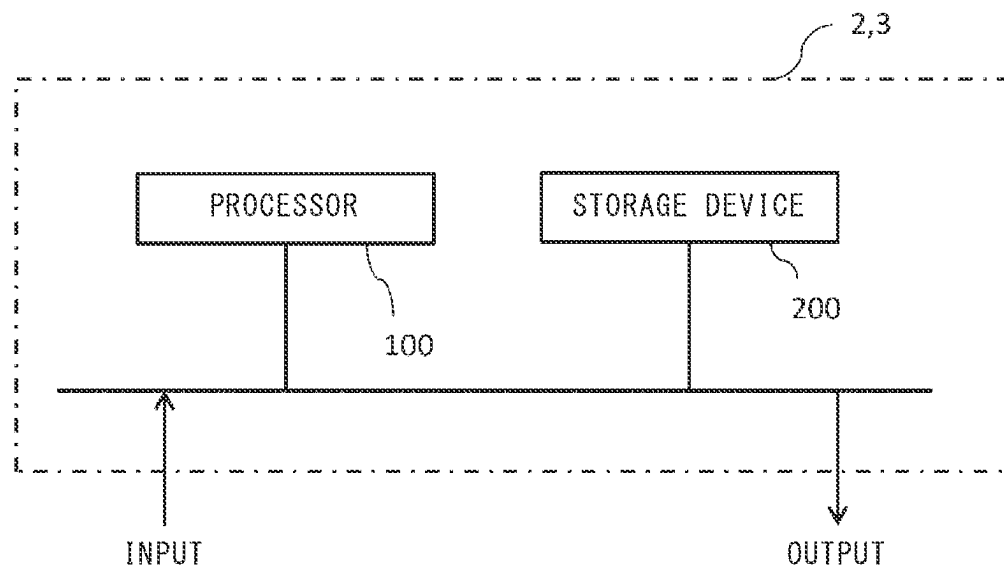


FIG. 7



ELECTRIC POWER STEERING DEVICE**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a National Stage of International Application No. PCT/JP2020/021270, filed May 29, 2020.

TECHNICAL FIELD

The present disclosure relates to an electric power steering device.

BACKGROUND ART

In a system that performs automatic parking of a vehicle, it has been known that, when a vehicle stops at a turnabout position and performs turnabout, static steering is performed to turn a steering wheel while the vehicle is stopped. Applying an excessive load on an Electric Power Steering (EPS), the static steering is likely to cause the EPS to be overheated. To prevent overheating damage of the EPS, currents need to be limited and overheating protection needs to be performed.

To solve these problems, there is a method of estimating the necessary number of static steering and, so as not to cause overheating protection to start unexpectedly during parking assistance, notifying a driver of the estimated number before the start of overheating protection (see Patent Document 1). Alternatively, after static steering has been performed and then the vehicle has been advanced, the steering direction is changed at a stage slightly earlier than when next static steering is performed. In this way, a static steering amount is reduced (see Patent Document 2). In addition, there is still another method of determining one route including static steering and another route not including static steering, and the static steering amount, depending on the temperature of the EPS (see Patent Document 3).

CITATION LIST**Patent Document**

Patent Document 1: Japanese Laid-Open Patent Publication No. 2009-19051

Patent Document 2: Japanese Laid-Open Patent Publication No. 2015-3565

Patent Document 3: Japanese Patent No. 6079596

SUMMARY OF THE INVENTION**Problems to be Solved by the Invention**

In a method disclosed in Patent Document 1, limitation or cancellation of parking assistance by overheating protection cannot be prevented.

In a method disclosed in Patent Document 2, after static steering has been performed and then the vehicle has been advanced, the direction is changed into the next steering direction at the stage earlier than usual, thereby reducing the steering angle for performing static steering. However, since a load state of a steering device is not detected, even if parking involving static steering by the steering device is sufficiently possible, a vehicle is parked through the route for reducing the static steering amount. Thus, the number of turnabouts until the completion of parking may be increased.

In a method disclosed in Patent Document 3, presence/absence of static steering is determined at the place for turnabout, on the basis of the temperature of the EPS. Accordingly, another route needs to be regenerated during parking control. The route different from the route initially set is to be used, thereby increasing the amount of steering until the completion of parking. Furthermore, a motor may be overheated due to the increased steering amount, and thus automatic parking may be interrupted.

The present disclosure has been made to solve the above problems, and an object of the present disclosure is to provide an electric power steering device that can avoid a steering disabled state and cancellation of parking assistance caused by an overheated motor during parking control and that allows parking control even in an overheating condition of the motor.

Solution to the Problems

An electric power steering device according to the present disclosure includes: route generation means for generating a route for driving from an own vehicle position to a target parking position; calculation command means for calculating a target steering angle command for causing a vehicle to drive along the route generated by the route generation means; an automatic steering control unit for calculating motor current for causing a steering angle to be formed according to the target steering angle command; an overheating protection current limit unit for calculating an overheating protection current limit value from the motor current; and a statically-steerable number determination unit for calculating a statically-steerable upper limit number on the basis of the overheating protection current limit value.

Effect of the Invention

Such an electric power steering device disclosed in the present disclosure can avoid a steering disabled state and cancellation of parking assistance caused by an overheated motor during parking control, and allows parking control even in an overheating condition of the motor.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic configuration diagram of an electric power steering device according to embodiment 1.

FIG. 2 is a flowchart illustrating operation of a statically-steerable number determination unit according to embodiment 1.

FIG. 3 is a schematic configuration diagram of an electric power steering device according to embodiment 2.

FIG. 4 is a flowchart illustrating behavior of a statically-steerable number determination unit according to embodiment 2.

FIG. 5 is a schematic configuration diagram of an electric power steering device according to embodiment 3.

FIG. 6 is another schematic configuration diagram of the electric power steering device according to embodiment 3.

FIG. 7 is an example of a hardware configuration diagram of the electric power steering device according to each of embodiments 1 to 3.

DESCRIPTION OF EMBODIMENTS

Hereinafter, an electric power steering device according to preferred embodiments for carrying out the present disclosure will be described in detail with reference to the

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drawings. In the drawings, the same reference characters denote the same or corresponding parts, and detailed description thereof will be omitted. Also in the subsequent embodiments, repeated description of components denoted by the same reference characters will be omitted in the same manner.

Embodiment 1

FIG. 1 is a schematic configuration diagram of an electric power steering device according to embodiment 1. The electric power steering device includes, as main components, a surrounding environment sensor group 1, a parking assistance device 2, an electric power steering (EPS) 3, and a steering angle sensor 4.

As the surrounding environment sensor group 1, a camera and sonar for monitoring the front, rear, or sides of a vehicle, a marker sensor for detecting markers in a parking lot, a GPS device for obtaining GPS information, a reception device for receiving map data in a parking lot, and the like are mounted in a vehicle. These sensors or devices detect condition information of the vehicle itself and surrounding information thereof. The information is transmitted to the parking assistance device 2 through a communication line of the vehicle such as a communication line of a Controller Area Network (CAN).

The parking assistance device 2 is a device for performing so-called automatic parking, and for outputting a command for changing vehicle behavior by using various information from the surrounding environment sensor group 1.

The parking assistance device 2 includes: route generation means 23 for receiving information from the surrounding environment sensor group 1, calculating own vehicle position information 21 and target parking position information 22, and generating a route to a target parking position; and calculation command means 24 for calculating a signal for a control command on the basis of output from the route generation means 23. Output from the calculation command means 24 is outputted to control the electric power steering (EPS) 3, a braking device 5, and a driving device 6.

In detail, the route generation means 23 detects, for example, a distance to a vehicle located to the front, rear, or side, lanes, a parking frame line, an obstacle, and the like, on the basis of information from the camera of the surrounding environment sensor group 1. From among the information, map information in a parking lot, OPS information, and the like, the route generation means 23 selects information necessary for performing parking control. On the basis of the selected information, the route generation means 23 calculates the own vehicle position information 21 and the target parking position information 22, and generates a route from an own vehicle position to a target parking position.

The calculation command means 24 calculates the target steering angle command 25 needed from the own vehicle position to the target parking position, according to route information generated by the route generation means 23, and outputs the target steering angle command 25 to the electric power steering (EPS) 3. In addition, the calculation command means 24 calculates and outputs an acceleration and deceleration command to the driving device 6 and a braking command to the braking device 5. For example, the calculation command means 24 issues commands, such as accelerating so as to travel along a route, steering to cause the vehicle to park between parking vehicles, braking wheels to cause the vehicle to stop at a vehicle stop position according to a route, and the like. That is, commands for performing vehicle speed control needed until the completion of parking

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are outputted to the electric power steering (EPS) 3, the braking device 5, and the driving device 6.

Furthermore, the calculation command means 24 calculates a target steering angle 26a and a target vehicle speed 26b needed from an own vehicle position to a target parking position. The difference between the target steering angle command 25 and the target steering angle 26a described above is as follows.

The target steering angle command 25 is a target steering angle needed for the calculation command means 24 to operate the vehicle according to the route generated by the route generation means 23, and is a command to be performed in real time. Meanwhile, the target steering angle 26a refers to all steering to be performed by the calculation command means 24 until the completion of parking along the route generated by the route generation means. The target steering angle 26a refers to all the steering, but also refers to reduced data thereof with contents limited to only specific points, such as the place where turnabout is performed, because the data amount for all the time-series data is enormous.

The electric power steering (EPS) 3 includes an automatic steering control unit 31, an overheat protection current limit unit 32, a statically-steerable number determination unit 33, and a motor control unit 34 that outputs motor current according to a target current.

The automatic steering control unit 31 receives the target steering angle command 25 from the parking assistance device 2 and steering angle information 41 from the steering angle sensor 4, and calculates a motor target current needed for steering according to the target steering angle.

The overheat protection current limit unit 32 limits the target current by using an overheat protection current limit value Y calculated from a motor current amount of the electric power steering (EPS) 3, to prevent damage due to overheat of the electric power steering (EPS) 3. That is, the overheat protection current limit value Y is a maximum current value of current that can flow through the motor at present, to prevent the motor from damaging itself.

With reference to a flowchart in FIG. 2, behavior of the statically-steerable number determination unit 33 of embodiment 1 will be described.

The statically-steerable number determination unit 33 estimates a target steering amount for each turnabout, from the target steering angle 26a and the target vehicle speed 26b inputted from the calculation command means 24 of the parking assistance device 2 and needed from an own vehicle position to a target parking position (step S1). Here, the target steering angle 26a to be inputted need not be all time-series data until the completion of parking assistance, and may be data only at each timing of performing turnabout with great change to the target steering angle.

On the basis of the target vehicle speed 26b, whether or not operation is static steering can be determined, and thus, the current amount necessary for turnabout can be estimated from the target steering amount, which is a variation amount of the target steering angle 26a (step S2). A predetermined estimated motor current amount per turnabout is corrected by using the estimated current amount (step S3). An overheat protection current limit value X to be used until the completion of parking assistance is calculated, from the corrected estimated motor current amount and the above overheat protection current limit value Y (step S4). This calculation is the same as that for the overheat protection current limit value Y to be calculated from the motor current amount in the overheat protection current limit unit 32, but is performed using the corrected estimated motor current

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amount instead of the motor current amount. That is, the
overheat protection current limit value X to be used until the
completion of parking assistance is a maximum current
value of current that can flow through the motor, calculated
from the estimated current amount until the completion of
parking. Since the motor is overheated when the current
flows, the overheat protection current limit value X is a
value that basically decreases depending on the corrected
estimated motor current amount.

When the overheat protection current limit value X until
the completion of parking assistance exceeds a predeter-
mined steerable determination current limit threshold (a
minimum motor current value needed until parking has been
completed) (step S5), a fact that steering can be performed
at the inputted target steering angle **26a** is outputted as a
determination result of a parking possibility determination
35b (step S6).

On the other hand, when the overheat protection current
limit value X is not greater than the steerable determination
current limit threshold, a fact that steering cannot be per-
formed at the inputted target steering angle is our as a result
of the parking possibility determination **35b** (step S7). Then,
a statically-steerable upper limit number **35a** is calculated,
from the overheat protection current limit value Y and a
predetermined estimated overheat protection current limit
value variation amount per turnabout with static steering
(step S8). Here, the statically-steerable upper limit number
35a is the number of times of turnabouts with static steering
allowable until a predetermined statically-steerable determi-
nation current limit threshold is reached. The statically-
steerable upper limit number **35a** is outputted to the route
generation means **23** of the parking assistance device **2** (step
S9).

In this case, the statically-steerable upper limit number
35a may be obtained through a map generated from a
relationship between the overheat protection current limit
value Y, the predetermined statically-steerable determina-
tion current limit threshold, and the estimated overheat
protection current limit value variation amount per turnabout
with static steering.

Alternatively, a gain depending on the target steering
angle variation amount inputted in advance is added to the
predetermined estimated overheat protection current limit
value variation amount per turnabout with static steering,
and the obtained value may be used for the calculation as
well.

In this case, whether or not a load of the electric power
steering (EPS) **3** in the route generated by the route genera-
tion means **23** is within an acceptable range can be checked,
thereby preventing automatic parking control from being
interrupted during automatic parking control due to the
electric power steering (EPS) **3** being in a high-load condi-
tion. Furthermore, even if it is determined that steering
cannot be performed, the statically-steerable upper limit
number **35a** is outputted to the route generation means **23**,
and thus the route generation means **23** can add the stati-
cally-steerable upper limit number **35a** as a condition to
regenerate a more feasible route.

According to the present embodiment, as described
above, whether or not the electric power steering (EPS) **3** is
in an overheat condition and whether or not the scheduled
parking assistance can be performed are determined. Even
when the electric power steering (EPS) **3** is in an overheat
condition, the statically-steerable number is determined in
consideration of a load of the electric power steering (EPS)
3, and thus the parking assistance device can be notified of
information necessary for calculation of a feasible parking

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route. Therefore, a shorter route in which parking is per-
formed than a parking route in which static steering is not
performed can be calculated, even when the motor is in an
overheat condition.

Embodiment 2

A configuration of an electric power steering device
according to embodiment 2 is basically not different from
that of embodiment 1, but input and output signals for the
parking assistance device **2** and the electric power steering
(EPS) **3** are different. FIG. **3** is a schematic configuration
diagram of the electric power steering device according to
embodiment 2. That is, in embodiment 1, control is per-
formed in a state in which steering to be performed has been
recognized in advance, whereas in embodiment 2, control is
performed in a state in which steering to be performed has
not been recognized in advance.

With such difference between the states to be a precon-
dition for control, the target steering angle **26a** and the target
vehicle speed **26b** needed from an own vehicle position to a
target parking position, which are outputted from the cal-
culation command means **24** to the statically-steerable num-
ber determination unit **33** in embodiment 1, need not be
outputted in embodiment 2. Furthermore, the parking pos-
sibility determination **35b** which is outputted from the
statically-steerable number determination unit **33** in embodi-
ment 1 need not be outputted in embodiment 2.

In embodiment 2, route generation means **23** detects, for
example, a distance to a vehicle located to the front, rear, or
side, lanes, a parking frame line, an obstacle, and the like, on
the basis of information from a camera of a surrounding
environment sensor group **1**. From among the information,
map information in a parking lot, GPS information, and the
like, the route generation means **23** selects information
necessary for performing parking control.

Furthermore, on the basis of the selected information, the
route generation means **23** calculates own vehicle position
information **21** and target parking position information **22**. A
statically-steerable upper limit number **35a**, which is out-
putted from the electric power steering (EPS) **3**, is added as
a condition, and as a route from an own vehicle position to
a target parking position, a parking route in which static
steering is performed within the statically-steerable upper
limit number **35a** is generated. In this case, turnabout in
which static steering is not performed may be included.

Calculation command means **24** calculates a target steer-
ing angle command **25** needed from the own vehicle posi-
tion to the target parking position, according to information,
generated by the route generation means **23**, on a parking
route in which static steering is performed, and outputs the
target steering angle command **25** to the electric power
steering (EPS) **3**. In addition, the calculation command
means **24** calculates and outputs an acceleration and decel-
eration command to a driving device **6** and a braking
command to a braking device **5**. For example, the calcula-
tion command means **24** issues commands, such as accel-
erating so as to travel along a route, steering to cause the
vehicle to park between parking vehicles, braking wheels to
cause the vehicle to stop at a vehicle stop position according
to a route, and the like.

As shown in a flowchart illustrated in FIG. **4**, a statically-
steerable number determination unit **33** calculates a stati-
cally-steerable upper limit number **35a**, from an overheat
protection current limit value Y calculated from a motor
current amount of the electric power steering (EPS) **3** in an
overheat protection current limit unit **32**, and a predeter-

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mined estimated overhear protection current limit value variation amount per turnabout with static steering (step S10). Here, the statically-steerable upper limit number 35a is the number of times of turnabout with static steering allowable until a predetermined statically-steerable determination current limit threshold is reached. The statically-steerable upper limit number 35a is outputted to the route generation means 23 of a parking assistance device 2 (step S11).

In this case, the statically-steerable upper limit number 35a may be obtained through a map generated from a relationship between the overhear protection current limit value Y, the predetermined statically-steerable determination current limit threshold, and the estimated overhear protection current limit value variation amount per turnabout with static steering.

In embodiment 2, a load condition of the electric power steering (EPS) 3 using a motor is not used at the time of generation of a route or calculation of commands, such as a target steering angle command, an acceleration and deceleration command, a braking command, and the like, each of which is processing of the parking assistance device 2. That is, since steering to be performed has not been recognized in advance, a load condition cannot be considered. Therefore, if the electric power steering (EPS) 3 falls into a steering disabled state during parking assistance due to overhear protection, it is necessary for a driver to take over the steering halfway through parking control. However, being in a high-load condition, the electric power steering (EPS) 3 cannot sufficiently perform steering assistance, that is, assistance to a driver's steering to be usually performed.

In this case, the statically steerable upper limit number 35a is sent from the electric power steering (EPS) 3 to the route generation means 23, whereby feasibility of driving assistance can be ensured. The electric power steering (EPS) 3 outputs, from the statically-steerable number determination unit 33, information on the number of times that the vehicle can perform turnabout with static steering during which the electric power steering (EPS) 3 may fall into a high-load condition, to the route generation means 23.

In this case, since the electric power steering (EPS) 3 notifies the route generation means 23 of the statically-steerable upper limit number 35a and a route is calculated before the start of parking operation, a route with a shorter vehicle movement distance can be generated, compared to a case where turnabout without static steering is performed after the electric power steering (EPS) 3 is detected to be in a high-load condition. For example, during parallel parking control in which a vehicle can perform static steering only twice due to the electric power steering (EPS) 3 being in a high load condition, steering without static steering is performed first, and then turning back with static steering is performed when a vehicle is driven between a front vehicle and a rear vehicle in parallel parking and when final turnabout is performed so as to align a parking direction between the vehicles within a limited vehicle-movable space. Thus, a route with a consequently shorter vehicle movement distance can be generated.

Embodiment 3

Embodiment 3 will be described with reference to FIG. 5 or FIG. 6. An electric power steering (EPS) 3 in embodiment 3 is different from embodiment 1 or embodiment 2 in that the electric power steering (EPS) 3 in embodiment 3 includes EPS temperature detection means, and a method for

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calculating an overhear protection current limit value in an overhear protection current limit unit 32 is also different as below.

In embodiment 1 or embodiment 2, the overhear protection current limit value is calculated solely from the motor current amount. However, the overhear protection current limit unit 32 of the electric power steering (EPS) according to embodiment 3 calculates the overhear protection current limit value from an EPS temperature from an EPS temperature detection means 36, and a motor current amount. This EPS temperature may indicate any of temperatures of heat generation components, such as a motor of the electric power steering (EPS) 3, a Field Effect Transistor (FET) mounted in an Electric Control Unit (ECU) of the electric power steering (EPS) 3, and the like, constituting the electric power steering (EPS) 3.

The EPS temperature detection means 36 enables more accurate calculation of the overhear protection current limit value according to temperature characteristics for each EPS component.

In FIG. 7, an example of a hardware configuration mounted in a parking assistance device 2 and the electric power steering (EPS) 3 is shown. The parking assistance device 2 and the electric power steering (EPS) 3 include a processor 100 and a storage device 200. Although not shown, the storage device includes a volatile storage device such as a random access memory and a non-volatile auxiliary storage device such as a flash memory. The storage device 200 may also include an auxiliary storage device of a hard disk instead of the flash memory. The processor 100 executes a program inputted from the storage device 200 to perform processing of, for example, the route generation means 23, the calculation command means 24, the automatic steering control unit 31, the statically-steerable number determination unit 33, or the like. In this case, the program is inputted into the processor 100 from the auxiliary storage device via the volatile storage device. The processor 100 may also output data such as a calculation result to the volatile storage device of the storage device 200, or may store data into the auxiliary storage device via the volatile storage device. A plurality of the processors 100 and the storage devices 200 may be provided.

Although the disclosure is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features, aspects and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described, but instead can be applied, alone or in various combinations to one or more of the embodiments of the disclosure. It is therefore understood that numerous modifications which have not been exemplified can be devised without departing from the scope of the present disclosure. For example, at least one of the constituent components may be modified, added, or eliminated. At least one of the constituent components mentioned in at least one of the preferred embodiments may be selected and combined with the constituent components mentioned in another preferred embodiment.

DESCRIPTION OF THE REFERENCE CHARACTERS

- 1 surrounding environment sensor group
- 2 parking assistance device
- 3 electric power steering (EPS)
- 4 steering angle sensor
- 23 route generation means

24 calculation command means
 25 target steering angle command
 26a target steering angle
 26b target vehicle speed
 31 automatic steering control unit
 32 overheat protection current limit unit
 33 statically-steerable number determination unit
 34 motor control unit
 35a statically-steerable upper limit number
 35b parking possibility determination
 36 EPS temperature detection means
 41 steering angle information

The invention claimed is:

1. An electric power steering device comprising:
 memory storing one or more instructions; and
 at least one processor configured to execute the one or
 more instructions to:
 generate a route for driving from an own vehicle
 position to a target parking position;
 calculate a target steering angle command to control a
 steering angle in real time to travel along the route;
 calculate motor current for causing the steering angle to
 follow the target steering angle command, based on
 steering angle data acquired from a steering angle
 sensor;
 calculate a first overheat protection current limit value
 based on the motor current;
 calculate a target steering angle for each of a plurality
 of turnabouts along the route generated for driving
 from the own vehicle position to the target parking
 position;
 estimate a current amount required for each of the
 plurality of turnabouts based on a variation in the
 target steering angles;
 correct a predetermined estimated motor current
 amount per turnabout, based on the estimate current
 amount, to obtain a corrected estimated motor cur-
 rent amount;
 calculate a second overheat protection current limit
 value until completion of parking assistance along
 the generated route, based on the corrected estimated
 motor current amount and the first overheat protec-
 tion current limit value;
 when the second overheat protection current limit value
 until the completion of the parking assistance is not
 greater than a predetermined statically-steerable
 determination current limit threshold, calculate a
 statically-steerable upper limit number, which is a
 maximum number of turnabouts permitted with
 static steering until the predetermined statically-
 steerable determination current limit threshold is
 reached, based on the first overheat protection cur-

rent limit value and a predetermined estimated over-
 heat protection current limit value variation amount
 per turnabout with static steering;
 generate an updated route for driving the own vehicle
 based on the statically-steerable upper limit number;
 and
 control the own vehicle to travel along the updated
 route.
 2. The electric power steering device according to claim
 1, wherein the first overheat protection current limit value
 represents a maximum current allowed to flow through a
 motor, and decreases based on the corrected estimated motor
 current amount, and
 wherein the predetermined statically-steerable determina-
 tion current limit threshold represents a minimum
 motor current value required to complete parking.
 3. The electric power steering device according to claim
 1, wherein
 when the overheat protection current limit value until the
 completion of the parking assistance exceeds the pre-
 determined statically-steerable determination current
 limit threshold, a determination result that steering is
 possible is outputted, and
 when the overheat protection current limit value is not
 greater than the predetermined steerable determination
 current limit threshold, a determination result that
 steering is impossible is outputted.
 4. The electric power steering device according to claim
 1, wherein
 the statically-steerable upper limit number is obtained
 through a map generated from a relationship between
 the overheat protection current limit value and the
 predetermined estimated overheat protection current
 limit value variation amount per turnabout with static
 steering.
 5. The electric power steering device according to claim
 1, wherein
 an EPS temperature detected by an EPS temperature
 detector is used in calculation of the overheat protec-
 tion current limit value.
 6. The electric power steering device according to claim
 3, wherein
 an EPS temperature detected by an EPS temperature
 detector is used in calculation of the overheat protec-
 tion current limit value.
 7. The electric power steering device according to claim
 4, wherein
 an EPS temperature detected by an EPS temperature
 detector is used in calculation of the overheat protec-
 tion current limit value.

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